

Yogesh. V

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Career Objective

To pursue a challenging career that offers me both technical task and leadership role thereby providing value addition to the organization.

Employment Details

Employer	Plintron Mobility Solutions Pvt Limited
Designation	Associate Technical Manager
Employment Start Date	23 November 2010
Employment End Date	12 September 2026
Total Work Experience	15 Years and 5 Months

Work Experience

- A result-oriented professional having excellent knowledge on Front End and Back End Automation and Load Testing of products.
- Experience in VoIP, VoLTE, 3G/4G/5G Core, SIP signalling, and multimedia services.
- Experience in developing Automation Frameworks – Cucumber Framework for front end applications and Robot Framework and Rest Assured Framework for backend applications.
- Experience in developing Flaskr Framework for Load Testing of Microservice based applications.
- Participation in Agile Scrum ceremonies (Sprint Planning, Daily Stand-ups, Sprint Reviews, Retrospectives).
- Collaborated with Product Owners and Developers for user story refinement and acceptance criteria definition.
- Used JIRA for sprint backlog tracking, test strategies, test plan, test case linking, and defect management.
- Experience in development infrastructure tools such as Jenkins, Git, Artifactory.
- Understanding and build Automated tests that fit into software delivery for building the CI/CD pipelines.
- Experience in code debugging and solving complex problems to build more reliable and scalable products.

Framework Experience

- Automation Testing (10+ years of experience).
- Robot and Flaskr Framework (10+ years of experience).
- Selenium, TestNG, Cucumber and Apache Poi (8+ years of experience).
- Postman, JMeter, Wiremock and Rest Assured (6+ years of experience).
- PyTest Framework and API testing (6+ years of experience).

Languages Known: Python, Java, JavaScript, Groovy and Html/CSS.

Tools Used: Soap UI and Xml Simulator; Postman, Wiremock and JMeter; MGTS for SS7, Seagull, Ixload and Sipp tool for IMS.

Database Used: Oracle (19c), SQL (2022), Redis and PostgreSQL.

IMS

- IMS Architecture & Protocols: Strong knowledge of IMS Core architecture (P/I/S-CSCF, HSS, PCRF/PCF, SBC, TAS, MRF) with hands-on expertise in SIP, SDP, Diameter, RTP/RTCP, TCP, UDP, SCTP, HTTP protocols.
- NFV / VNF Deployments: Deployed and managed VNF-based IMS components (P/I/S-CSCF, PCRF, HSS) on OpenStack NFV environment, optimizing CPU, memory, and storage resources by tuning VM Flavors and configurations.

- **Functional & Load Testing:** Executed VNF functional and load testing using Seagull, Spirent, IXIA, Sipp, ensuring 3GPP standards compliance for VoLTE and IMS products.
- **Diameter & VoLTE Validation:** Validated Diameter signalling (Rx, Gx interfaces) for PCSCF-PCRF applications, emergency call handling, roaming scenarios, and CS network interworking.
- **Automation Framework Development:** Built and enhanced Robot Framework test suites for OCS/NSS/IMS/REST/XML-based products, automating regression for 30+ telecom products. Migrated framework to Robot Framework 7 for higher stability and performance.
- **Network Tracing & Debugging:** Skilled in capturing and analysing traces using tcpdump, tshark, Wireshark, troubleshooting SIP/IMS signalling and RTP flows.
- **Implemented automated testing of LocalAI with Llama-2-7b-chat local LLM** using Robot Framework, enabling cost-effective evaluation of open-source chat models with OpenAI-compatible API without cloud dependencies.
- **Load & Stress Testing:** Prepared and executed high-scale load scenarios with IxLoad, Seagull, Sipp, delivering test results to improve IMS system reliability under heavy traffic.
- **KPI Optimization & Collaboration:** Worked with network teams to optimize QoS & KPIs such as Call Setup Success Rate (CSSR), Call Drop Rate (CDR), Registration Success Rate, RTP Jitter/Latency, improving system performance.
- **Architecture Review & Improvements:** Provided architecture-level reviews for IMS/Telecom systems, recommending design optimizations for performance, reliability, and scalability.

5G (SBA) and HLR, HSS Products

- **5G Core (SBA) Expertise:** In-depth knowledge of UDM, UDR, AUSF, PCF, AMF, SMF, UPF nodes and 5G-AKA authentication with Service-Based Architecture (SBA).
- **UDM / UDR Validation:** Validated subscriber profile management, policy retrieval, and session continuity in 5G Core through UDM-UDR functions.
- **Authentication Testing:** Executed AUSF-UDM authentication using 5G-AKA and EAP-AKA flows, ensuring seamless interworking with LTE HSS during VoLTE-VoNR calls.
- **SBA API Testing:** Performed validation of SBA interfaces (UDM, UDR, AUSF) via HTTP2/JSON APIs, troubleshooting signalling failures with Wireshark for registration/authentication issues.
- **Automation:** Automated UDM/UDR/AUSF test scenarios using Python + REST API scripts, reducing manual validation effort and improving coverage.
- **HLR / HSS Expertise:** Hands-on experience with HLR and HSS applications using MAP, CAP, and Diameter protocols, including subscriber provisioning, authentication, and mobility management.
- **Diameter Signalling & Scenarios:** Validated Cx, Sh, S6a, Swx interfaces for VoLTE/VoWiFi services. Designed and executed Diameter procedures (AIR, ULR, UAR, MAR, SAR, LIR, RTR) with Seagull for HSS functional & load testing.
- **Telecom Protocols Knowledge:** Strong understanding of GSM, SS7, SIGTRAN, and Diameter nodes, enabling smooth interworking between circuit-switched and packet-switched networks.
- **KPI Monitoring:** Delivered key metrics such as Registration Success Rate, Authentication Latency, Policy Retrieval Accuracy for performance benchmarking.
- **Subscriber Data Platforms:** Worked on NDS (Network Data Storage) based subscriber data management in HSS/UDM for LTE/IMS/5G environments. Validated SDL integration with HSS, PCRF, and OCS for unified subscriber data storage and access.

CPaaS (Voice/Messaging Application)

- **Containerization & Orchestration:** Designed and deployed Dockerized test frameworks (Selenium, Rest Assured, Pytest) on Kubernetes clusters for scalable automated testing.
- **CI/CD Pipeline Automation:** Executed automated test suites inside Kubernetes pods as part of CI/CD pipelines (Jenkins, Git, Ansible) ensuring seamless integration and faster feedback cycles.
- **Dynamic Test Environments:** Configured Kubernetes namespaces dynamically for parallel execution of test environments, reducing test execution time.

- **Test Reporting:** Integrated automated test results with Allure Reports, persisted reports in K8s volumes, and enabled test trend analysis.
- **Robot Framework + Selenium:** Developed Robot Framework test suites with Selenium WebDriver for web UI validation, enabling faster regression cycles.
- **API Testing & Microservices:** Enhanced Pytest-based Rest Assured framework for REST API validation of microservices, including custom assertion handling.
- **Data Management & Load Scripts:** Developed Python-based load scripts for Redis DB operations (record insertion, deletion, count validation) to support large-scale microservice testing.
- **Config Parsing & Data Validation:** Automated reading of config files, parsing JSON/XML API responses, and transforming/validating data between microservices.
- **Performance Monitoring:** Enhanced Flaskr-based Python framework to capture Docker server metrics (CPU, memory) during load testing for performance optimization.
- **Network Monitoring:** Implemented real-time network monitoring using Python integrated with SNMP, syslog, REST APIs, and visualized metrics for proactive alerting.
- **Version Control & Collaboration:** Maintained automation scripts in Git repositories with proper branching strategies, pull requests, and peer reviews for collaborative development.

Data Analysis and Networking

- Built a lightweight Flaskr framework with knowledge of HTML, CSS, JavaScript, and Python for backend product load testing and monitoring dashboards.
- Utilized Pandas to clean, transform, and analyze large datasets (CSV, Excel, SQL), improving data processing efficiency by 60%; leveraged backend application statistics for trend analysis.
- Designed custom ETL pipelines using Pandas for filtering, merging, aggregation, and pivot table generation.
- Leveraged NumPy arrays for high-performance numerical operations, reducing dataset processing time by up to 50%.
- Automated daily/weekly reporting by integrating Pandas with Excel/CSV/SQL outputs, enabling faster business decision-making.
- Enhanced Python scripts with Pandas & NumPy to validate and reconcile large datasets across systems, ensuring accuracy.
- Applied OOPS concepts (Abstraction, Encapsulation, Method Overloading/Overriding) in automation frameworks for modularity and scalability.
- Built automated test cases for REST APIs, including TCP-level timeout and retry handling for robustness.
- Improved Python scripts to debug packet loss during audio exchange in IMS applications and troubleshoot transport-layer latency issues.
- Developed systematized troubleshooting scripts to detect network anomalies (high CPU, flapping interfaces) and trigger proactive alerts via email.

Chatbot

- Built 50+ pytest test cases with parameterization, fixtures, and mocking for external API calls.
- Used mocking for external LLM API calls to reduce costs. Parameterized tests for multiple query variations.
- Validated FAISS vector store retrieval quality – Tested semantic search: compared queries against document embeddings.
- **RAG Quality Metrics:** Implemented RAGAS framework metrics.
- **Context Recall:** Measured retrieval completeness (% of relevant documents found by system) – Target: 90%+.
- **Context Precision:** Measured retrieval accuracy (% of retrieved documents that were relevant) – Target: 85%+.
- **Combined metric validation:** Ensured system finds right documents and doesn't over-retrieve noise.
- **LLM Evaluation:** Implemented custom metrics for evaluating response quality – Faithfulness: Verified outputs grounded in retrieved source documents (scored 0–1); Relevance: Measured answer relevance to user queries using semantic similarity; Hallucination Detection: Flagged claims unsupported by source documents.

- Performance Testing: Validated end-to-end latency (retrieval + generation), token usage, and cost per query.
- CI/CD: Integrated pytest into GitHub Actions with automated metric reporting and regression detection.
- Results – Context Recall: 92% (comprehensive document retrieval); Context Precision: 88% (minimal irrelevant results); Faithfulness: 96% across 10,000+ document corpus; Hallucination rate reduced from 20% to 5%; Testing efficiency: 40% reduction in QA cycle time; Quality gates: 12 critical bugs detected pre-production; Reliability improved from 85% to 98% uptime.

CRM and Mobile Number Portability

- Strong experience in Customer Care (CRM, Reseller, Retailer apps) and MNP applications, with a mix of Manual (30%) and Automation Testing (70%).
- Hands-on with SQL queries, stored procedures, debugging, and validating backend data across telecom applications.
- Designed and developed Cucumber Automation Framework for CRM, Reseller, and Retailer systems.
- Automated test cases using TestNG, JUnit, and Cucumber, covering both functional and regression testing.
- Proficient in testing Web Services, Win Services, SOAP, XML, and HTML-based applications.
- Good understanding of Web Servers, Application Servers, and Database Servers for troubleshooting and validation.
- Successfully led a team of 10 members in both Manual and Automation Testing.
- Recognized with an award for highest number of defects identified (5000+) across 3 years, driving major quality improvements in products.
- Developed data-driven Rest Assured automation framework for OCS Application interacting with microservices.
- Implemented parallel execution using TestNG with a multi-threaded environment in Rest Assured to optimize test execution.
- Utilized Wire-Mock to simulate APIs (REST/XML), enabling both automation and performance testing.
- Integrated JMeter and Robot Framework for automation of a Postpaid application with REST API architecture.
- Analysed Java application stability by capturing and reviewing thread dumps & heap dumps to detect memory leaks.

Education

Qualification	Institution	Board / University	Year	Score
M.Tech (AI & ML)	BITS Pilani (WILP Program)	BITS Pilani	2026	Currently pursuing
B.E. (ECE)	GKM College of Engineering, Chennai	Anna University	2009	70%
12th Standard	Zion Matric Higher Secondary School, Chennai	State Board (Tamil Nadu)	2005	93%
10th Standard	Zion Matric Higher Secondary School, Chennai	State Board (Tamil Nadu)	2003	88%

Personal Profile

- Name: Yogesh. V.
- Father's Name: P. Venkatachalam
- Date of Birth: 11 March 1988
- Marital Status: Married
- Gender: Male
- Languages Known: English, Hindi, Tamil

Declaration

I hereby declare that the above-mentioned information is correct up to my knowledge and I bear the responsibility for the correctness of the above-mentioned particulars.

Place: Chennai

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